C05 # Memory Stack (4 levels)

Design Intermediate

Where the information Claude uses lives and how to manage it

The 4 memory levels

Claude Code saves local transcripts and enables auto-memory by default. Use explicit instructions and inspect stored memory to decide which information should influence future work.

Level 1: CLAUDE.md (explicit rules) Markdown files read at the start of each session. They contain project conventions, important commands, and architecture decisions. Human-readable, versioned with the code.

```
~/.claude/CLAUDE.md        # Global (all sessions)
.claude/CLAUDE.md          # Project (team)
CLAUDE.md                  # Root (current folder)
```

Level 2: Auto-memories (automatic learning) Claude automatically records decisions made during sessions in `~/.claude/projects/*/memory/MEMORY.md`. This mechanism is transparent: you can review these files to audit what Claude has learned.

Level 3: Tasks API (structured cross-session state) For multi-step workflows, opt into task tools with `CLAUDE_CODE_ENABLE_TODO_TOOLS=1`. Use `CLAUDE_CODE_TASK_LIST_ID` to select a persistent list shared across sessions, and inspect its current records before resuming work.

Level 4: Agent memory field (local context) Each specialized agent can have its own declarative memory in its YAML frontmatter, independent from the main session.

What disappears each session

The live context is session-specific, but saved conversations can be resumed. `claude --continue` selects the recent conversation; `claude --resume` lets you choose an older one. Neither restores the previous filesystem or remote-service state.

Selection rule

Need	Level to use
Permanent project convention	CLAUDE.md
Decision made in session	Auto-memory (or manual CLAUDE.md)
State of a long task	Tasks API
Context of a specialized agent	Agent memory field
Ephemeral configuration	Session (no persistence)

CLAUDE.md as composed memory

> “You should never have to correct Claude twice for the same mistake.” (Boris Cherny, creator of Claude Code)

Each mistake corrected once becomes a CLAUDE.md rule. Over 3 months, a well-maintained CLAUDE.md accumulates the conventions, gotchas, and architecture decisions that accelerate all subsequent sessions, including for new team members.

Review new rules for relevance, remove duplicates and stale decisions, and verify that the instructions improve the tasks they target.

Target size: Keep instructions concise, with roughly 200 lines as a practical starting point. File length is not a measured threshold for model coherence.